

Problem A: Cygnid Cipher

Group35 problem **cygnidcypher** - Time limit: 2s

The Grand Emperor Cygnus has given the command, the Black Swans of the Derbarl Yerrigan shall wage all-out war against the Mallard Ducks for control of the lake's prime bread-crust foraging zones. To transmit the strike timing across the river without duck patrols intercepting the message. The swans send their times through a series of squawks (0) and squeeks (1). Find the minimum number of squawks and squeeks that a swan must make in order to get their message across

Input

The input consists of a single line containing a non-empty string S ($1 \leq |S| \leq 10^5$), representing the secret battle strategy. S contains only lowercase English letters (a-z) and spaces.

Output

Output a single integer representing the total bit length of the encoded message.

Sample Tests

Sample Input 1

attack ducks at dawn

Sample Output 1

60

Sample Output 2

quack no more

Sample Output 2

42
